BrickPlan: A Benchmark for Visual Assembly and Error Recovery

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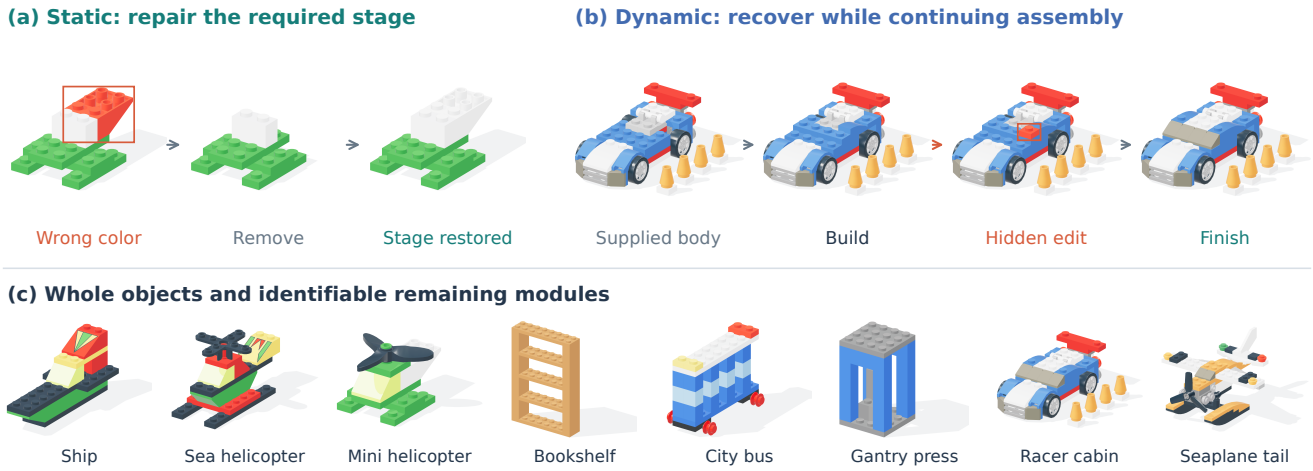


Figure 1. **Visual assembly requires both correction and continued progress.** (a) Static repair restores a supplied structure to its stated stage. (b) Dynamic assembly continues through an unannounced change, with the goal and inspection tools always available. (c) Tasks span complete small objects and named modules supplied with their surrounding bodies. The main collection contains 3,697 static and 1,755 dynamic candidates over 50 whole-object and 13 module configurations. Coral callouts explain the changed pieces. Scenes are exact-state publication renders, not model results.

Abstract

Assembly agents must maintain and repair correct work while progressing toward a goal. We introduce BRICKPLAN*, a benchmark connecting static structure repair with continuous assembly under hidden interventions. Its 5,452 main candidates cover local structures, complete objects, and identifiable remaining modules. Seven physical error labels are separated from access-related rework and repeated incidents. Agents inspect RGB states and manipulate one brick at a time; terminal scoring accepts interchangeable parts and alternative legal construction orders. We define correction, damage, and recovery metrics with explicit exposure and control requirements, preventing an unreached fault or failed API call from becoming a visual-miss claim. Reference replay and route diagnostics validate execution; formal learned-agent and independent human results remain unmeasured.*

1. Introduction

An agent assembling a car prepares its roof, but an earlier piece is removed before the next observation. The target remains unchanged; completed work does not. Finishing requires relating the goal, action history, and actual scene, then executing a legal repair. Choosing the next step on an undisturbed manual does not test this ability.

Such a task also exposes weaknesses in evaluation. A car can be assembled in several legal orders. Comparing each intermediate state with a single demonstration rejects legitimate progress. Scheduling a fault only when the agent reproduces that demonstration's prefix can prevent a valid alternative route from ever encountering the fault. Conversely, a model that cannot build normally provides little evidence about repair under intervention. Completion, physical legality, exposure, and recovery need different definitions and denominators.

Visual specification also matters. A finished-object image can hide joints; an arbitrary crop can lack identifiable purpose. BRICKPLAN separates local static diagnostics

from dynamic object construction. Small original LDraw files retain every part; large scenes supply a recognizable body and a named remaining module. One target defines acceptance; available manuals explain structure without prescribing its only valid sequence.

Prior work establishes buildable brick generation [1, 5], manual execution [10], step correction [9], and interactive reconstruction [8]. Visual-spatial benchmarks further show the value of separating task families and reasoning demands [11, 12]. Our focus is continuous error recovery under a controlled visual action contract: the world can change after the agent has already made progress, and the agent is not interrupted to answer an anomaly questionnaire.

We contribute: (1) a source-organized collection of 5,720 distinct task records, with 5,452 main candidates and separately retained diagnostics; (2) a shared RGB/action contract that accepts physical endpoint equivalence and legal alternative orders; and (3) metrics distinguishing static correction from damage and dynamic recovery from exposure, with reference checks and reproducible protocol diagnostics. The current evidence establishes executor properties. The planned agent study will test perception and planning under this contract.

2. Related Work

Generation, execution, and reconstruction. BrickGPT generates buildable structures with validity checks and rollback [5]. BrickNet represents diverse LDraw parts through connection graphs and executable sequences [1]; BrickBench studies agentic brick design [2]. Our intended structure is supplied: the objective is restoration or completion rather than open-ended design quality.

MEPNet translates visual manuals into executable plans [10]. LEGO-ECA and SCANet study errors associated with assembly steps [9]. Break and Make combines interactive inspection, disassembly, and reconstruction [8]. Our static condition connects to correction; our dynamic condition retains the ongoing agent history across an external change.

Spatial reasoning and adaptation. LEGO-Puzzles and PhyBlock study spatial understanding and block planning [4, 6]. VSI-Bench evaluates spatial recall from video [12], while Spatial457 organizes visual questions by spatial capabilities [11]. AsgardBench and RoboRecover address interactive adaptation and recovery [3, 7]. Thus neither spatial reasoning nor recovery itself is new. Our setting combines a specified brick target, one-part physical actions, alternative-order acceptance, and qualified comparison of control and intervened continuations.

Table 1. **Related task contracts.** Inputs and responses differ, so scores are not directly comparable. RGB includes ordinary task text. Our manuals are available only on supported tracks.

Setting	Goal / evidence	Response
BrickGPT	Text design request	Build sequence
BrickNet	Unconditional / text	Build sequence
VSI-Bench	Video + question	Spatial answer
Spatial457	Image + question	Spatial answer
BRICKPLAN static	Stage goal + current RGB	Legal repair actions
BRICKPLAN dynamic	Goal + RGB history; manual	Continue, repair, finish

3. The BrickPlan Collection

3.1. Task scopes and construction

Table 2 summarizes the main scopes. Static repair starts from a supplied state and ends at a specified stage. A local structure can be useful even when it is not a recognizable complete object. Dynamic tasks instead require a complete small object or an identifiable remaining module. Both retain a fixed coordinate frame and white-baseplate environment; module tasks additionally supply an immutable surrounding body.

The dynamic collection has 63 operation configurations. Fifty are whole objects: all 42 earlier procedural designs, five longer procedural objects, and three complete original LDraw files. The originals are a ship (4018, 12 parts), sea helicopter (4017, 13), and mini helicopter (4037, nine). Their source instances, colors, and poses are retained without dropping parts. This denotes source-file completeness, not retail bill-of-materials verification. Longer builds include a bookshelf, bus, bridge, gantry press, and cottage with 21–32 active pieces.

Thirteen module configurations come from twelve community scenes, plus an enlarged museum roof on an existing source. Their supplied bodies make the purpose of the remaining work explicit. The original modules have 5–12 active pieces; the enlarged roof has sixteen. A large source scene therefore does not imply a large action task. All earlier chairs, tables, cars, and mechanisms retain their provenance; the supplement illustrates every main dynamic configuration and lists all 160 original static configurations.

3.2. Physical errors and process demands

Figure 2 shows seven basic physical labels: missing, extra, wrong color, wrong part, position/height offset, non-equivalent orientation, and positional swap. Wrong-part labels include size and thickness. Extra-layer examples add

Table 2. **Dataset scopes and task types.** Main candidates sum to 5,452. S0/S1 denote correct-stage acceptance/static repair; D0/D1/D2 denote normal construction/one incident/repeated incidents. Counts include normal controls but scores keep them separate.

Scope	Total	S0	S1	D0	D1	D2	Start and endpoint
Local static structure	3,315	173	3,142	–	–	–	Supplied local state → required stage
Whole-object static stage	148	38	110	–	–	–	Object stage → stage target
Named-module static stage	234	49	185	–	–	–	Body + module state → stage target
Whole-object dynamic	1,534	–	–	50	1,464	20	Empty plate → complete object
Named-module dynamic	221	–	–	13	185	23	Supplied body → completed module
Main candidates	5,452	260	3,437	63	1,649	43	Scores remain scope-specific

local pieces; they do not duplicate an entire object layer. Labels may co-occur, so a compound state is not forced into one exclusive class.

Process tags capture a different burden. Premature covering can obstruct later insertion; buried repair can require temporarily removing correct work. A missing repeated element still carries the missing-piece label. Figure 3 shows why repair complexity is not exhausted by counting defective parts. Several simultaneous defects also differ from separate accidents over time: the main set has 37 two-event and six three-event conditions.

Twelve approved human-inspired source tags motivate these synthetic conditions. They are analogues, not a dataset of observed human mistake trajectories or estimates of human error frequencies. Excluded and deferred categories remain documented in the release specification.

3.3. Structural descriptors and source grouping

We record build length B1 ($\leq$ 12 active target pieces), B2 (13–24), and B3 ($\geq$ 25). Static B counts the acceptance stage, not future manual pieces or supplied scenery. Rework R0 denotes controls; R1 requires at most two reference moves without correct-cover removal; R3 includes such removal or at least seven reference moves; R2 contains the remaining ordinary/multiple-defect rework. These are witness burdens, not proven minimum costs. T0–T3 count external events; static errors already exist at the start and are T0.

Provisional grades combine the descriptors: G3 means B3, T3, or repeated events with R3; otherwise B2, R2/R3, or T2 gives G2; the remainder is G1. Their main counts are 2,577/2,823/52. They are planning strata rather than empirically calibrated easy/medium/hard labels. Visibility remains an independent admission field.

Original sets/procedural families define 154 disjoint source groups; 45 displayed/debugged sources are public development. The existing-source test has 392 main candidates (309 local static, 83 whole dynamic). Community modules and original-object examples are currently public development; new-source claims require untouched sources.

4. Interactive Task and Execution

4.1. A unique target, RGB observations, and one action

A task specifies fixed context B, movable parts P, registered poses C_p, initial state s_0, and acceptance target g. At each decision the agent sees the actual current RGB state and can inspect supported isometric, rear, or top views. Previous observations provide temporal evidence. A static target is its required stage; a dynamic target is the completed object/module. Future manual pages remain references and never silently replace the acceptance target.

Current/target panels share framing (800×600 each on a 1600×636 canvas). Modules retain full-object context. Supported tracks offer on-demand manuals; legacy local static does not. RGB, neutral rules, and anonymous handles are public. Inventory lists spares, not live remaining counts; geometric state stays private.

The shared response is a single direct action object:

$$a_t \in \{\mathrm{attach}(p, c), \mathrm{detach}(p), \mathrm{inspect}(q), \mathrm{finish}\}. \quad (1)$$

A manipulation moves exactly one brick. Inspection can query a view, part, candidate preview, inventory page, or supported manual step. Previews are hypothetical; they neither execute a placement nor certify it as correct. The current input version uses the same response envelope across tracks. Free-coordinate prediction and explicit text-state assistance are separately planned experiments with different information/action burdens.

4.2. Legal transitions and endpoint equivalence

Attachment checks a supported compatible connection, endpoint collision, and straight insertion path; detachment checks extraction. A rejected attempt leaves geometry unchanged. Legality feedback can aid planning, but a legal placement may still be semantically wrong. Subsequent correction can still yield success. Finite poses and rigid geometry do not model force, slipping, articulated motion, or a gripper.



Figure 2. **Physical discrepancies versus a correct control.** Each cell pairs the actual current state with its unique stage target. E01–E06 use a mini-helicopter stage, E07 uses a bus, and S0 needs no modification. Matched cameras make the differences interpretable; coral outlines are paper-only difference callouts. E04 magnifies thickness; E05/E06 use rear views. Geometry remains certificate-exact.

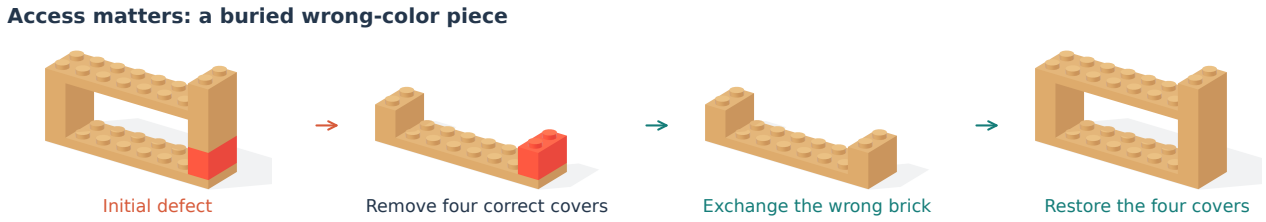


Figure 3. **One wrong piece can entail substantial correct-work removal.** A buried wrong-color piece in a seven-piece bookshelf stage is repaired by a verified ten-action witness: remove four correct covers, detach the wrong piece, attach its correct counterpart, and restore the covers. The final configuration, rather than these particular intermediate states, defines success. The witness is feasible, not proved shortest.

Let $\sigma(s)$ be the multiset of installed part shape, color, and pose, using only supported symmetry equivalences. Completion is

$$C(\tau) = \mathbf{1}[a_T = \text{finish}]\mathbf{1}[\sigma(s_T) = \sigma(g)]. \tag{2}$$

This accepts interchangeable pieces and alternative legal orders, with only justified part symmetries and no arbitrary global alignment. Immutable context is excluded from scores.

4.3. Intervening on actual progress

Hidden edits are applied after accepted manipulations and before the next observation. The private log retains their timing, legal primitives, and realized effect. Adaptive module selectors search the actual installed state; older conditions can have template-dependent opportunities. An infeasible edit is deferred or remains untriggered. A scheduled event or nominal step threshold is not evidence of exposure.

Multiple incidents each receive a separate record. Removing a just-added piece can cancel the latest action and leave before/after images alike; this action-outcome case is flagged separately from a change to older work. An edit that removes an already-wrong placement may help the agent. Premature covering can block future access even when the cover belongs to the final goal. These cases require effect checks, rather than judging all incidents solely by added final-state residual.

5. Evaluation: Correction, Exposure, and Recovery

5.1. Static repair without collateral damage

S0 denotes already-correct acceptance (260 tasks); S1 denotes repair (3,437). Let I, F, G be initial, final, and target movable-placement multisets and $H = I \cap G$ the initially correct work. Exact repair SR averages Eq. 2 over valid S1 runs. Partial quality, error reduction, and final damage are

$$\text{F1}_{\text{place}} = \frac{2|F \cap G|}{|F| + |G|}, \tag{3}$$

$$d(A, G) = |A - G| + |G - A|, \tag{4}$$

$$\text{ER} = 1 - \frac{d(F, G)}{d(I, G)}, \qquad \text{DR} = \frac{|H - F|}{|H|}. \tag{5}$$

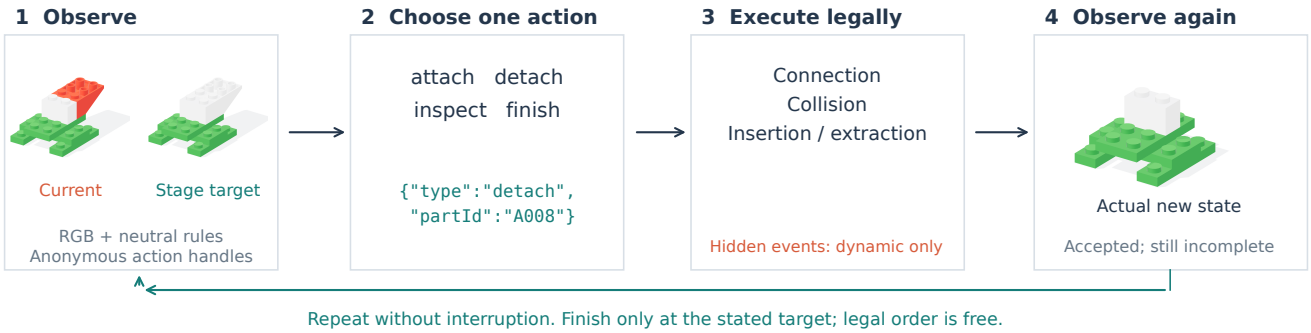


Figure 4. **Concrete observation–action–execution loop.** An example static repair removes the wrong-color brick. The resulting state is legal but incomplete, so the next decision must use the new observation. Each response is one direct action object; inspections remain available throughout. Dynamic events occur between actions without an error announcement. Private fault labels, raw pose maps, and reference repairs never enter the public observation. States are re-rendered here for print; agent inputs use the native RGB canvases.



Figure 5. **Partial progress is not completion.** In this six-piece stage, removing the wrong-color brick reduces residual from two to one and increases F1, but the task remains unfinished. Replacing it and finishing gives SR one. These are exact scorer values on reference states, not model scores.

ER can be negative. Residual counts physical mismatches, not human mistakes or proven shortest repair actions. Temporarily removing and restoring a correct cover does not create final DR. On S0, false repair FR records any accepted attach/detach, including remove-and-restore; final acceptance is also reported. S0 and S1 are not pooled. Undefined denominators are N/A; F1 of two empty sets is one.

Figure 5 illustrates why SR and partial quality are both needed. Supplemental scores track restoration of original missing placements, clearance of original excess, and attribute-combination F1. A connection-adjacency F1 is planned but is not yet uniformly implemented.

5.2. Dynamic completion needs qualified exposure

D0 measures normal construction (63 tasks); D1 assigns one incident (1,649); D2 assigns repeated incidents (43). BCR is valid D0 completion. ICR is raw completion of valid intervention-assigned runs, including untriggered ones. TR counts applied events over planned events. Neither raw completion nor trigger coverage establishes visual detection.

A paired study executes a common prefix once, saves the environment, conversation, images, mapping, and used budget, then branches into normal and fault continuations (Fig. 6). Subsequent legal orders can differ. Define Q as pairs with valid branches, a verified shared prefix, all scheduled events applied, and verified incident effects. For normal/fault success c_i, f_i, report

$$\mathrm{CRR} = \frac{\sum_{i\in Q} c_i f_i}{\sum_{i\in Q} c_i}, \qquad \Delta\mathrm{SR} = \frac{1}{|Q|}\sum_{i\in Q}(c_i - f_i). \quad (6)$$

CRR additionally conditions on successful normal construction; the signed paired difference retains normal failures. Report planned/valid pairs, failed controls, untriggered/partial exposure, unverified effects, and eligible N. Two fresh runs sharing a nominal seed are not proven shared-prefix pairs. An unreached fault is not a missed detection.

5.3. Repair while making new progress

For an event with states B before and A after, define lost correct placements $L = (B \cap G) - A$ and incremental excess $X = (A - G) - (B - G)$. Recovery restores these quantities, without requiring the later state to equal B. Additional correct construction is allowed. Each event window ends before the next external intervention.

Report first-recovery decisions, window-end recovery, and retention over at least two follow-up decisions. Inspections and invalid actions count as decisions. Observed recurrence fails retention; a correct but too-short window is N/A. A pure access obstruction with empty L, X receives no automatic zero latency. All-event D2 recovery requires complete effective exposure and final success; partial exposure remains a separate outcome. Repair@2/4/8 is reserved

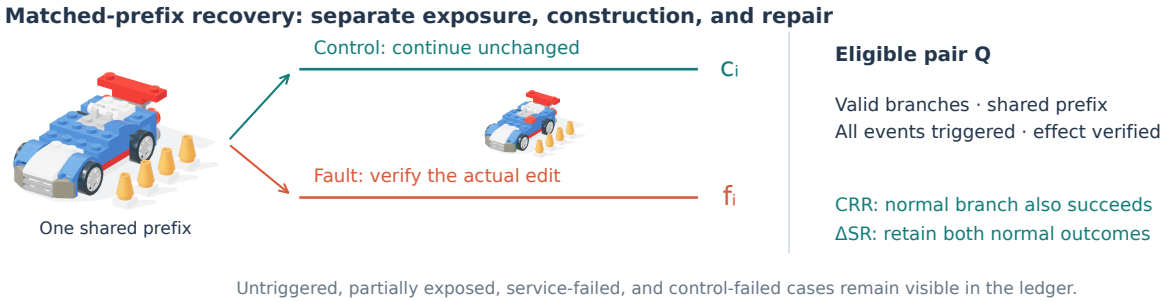


Figure 6. **Recovery and paired effect use different populations.** Technical/event gates define Q; CRR further requires a successful normal branch. ΔSR retains both normal outcomes. Thus weak construction, incomplete exposure, and poor repair remain distinguishable. The pictured states illustrate a shared module prefix and a reference event; branch outcomes are variables, not measured model results.

for sufficiently observed single-event/checkpoint windows. A future checkpoint C bank supplies the prefix explicitly and will not count that prefix as model-built progress.

5.4. Validity, cost, and aggregation

Wrong finish and model budget exhaustion are ability failures. Final service errors, harness errors, and external stops are censored from ability rates and reported separately. A recovered same-decision retry can produce a valid run, while its charge remains. Parsing, schema/handle errors, and physical rejections have separate counters. An incorrect episode alone cannot establish a visual detection miss; independent snapshot detection/localization remains a separate planned task.

Normalize canonical-task weights, average within objects, and report object- and source-macro estimates alongside task-micro references. Keep input version, scope, and task type separate. Report valid coverage and completion per planned attempt so access failures cannot disappear. Costs include inspections, all attempts, images, tokens, shared prefixes, both continuations, and billed retries. Reference-cost ratios use feasible witnesses and can be below one; they are not optimality or SPL. Future source-clustered intervals retain variants together. One run per condition does not establish repeatability.

6. Dataset Analysis and Protocol Diagnostics

6.1. Coverage and execution checks

Figure 7 separates diversity of tasks, operation length, and error coverage. Whole-object configurations span longer builds than most named modules, while main G3 has only 52 candidates. Error balance is uneven: color and missing-piece conditions are frequent; positional swaps are sparse and absent from the module-dynamic scope. Multilabel coverage is therefore not evidence of equal support for every claim.

Physical deduplication leaves 5,720 distinct records. The main set excludes 268 historical dynamic diagnostics; retained aliases count as execution conditions, not additional tasks. The supplement details this accounting.

Fresh reference replay passes all 5,922 retained execution conditions with zero failures. This is a solvability/execution audit, not a model success rate. The three original objects have 36 faulty reference conditions meeting a 128-pixel change threshold at 24 RGB levels in at least one native view, plus ten controls. Adaptive locations need their own check. Old cropped-module checks do not certify the new full-context interface. Complete image/human qualification remains outstanding.

6.2. Does reference-order scheduling change the test?

A selected-route diagnostic uses twelve community modules, two legal routes, and four conditions per route: 96 trials. Alternative routes were chosen as regression counterexamples, not sampled from a model. All 24 normal routes reach the same targets. Requiring every manual prefix accepts zero of twelve alternatives, demonstrating why terminal physical equivalence is necessary.

Exact-prefix opportunities cover 9/36 alternative-route faults, versus 36/36 actual-state events (Fig. 8). A blind policy continuing its normal route finishes 24/24 controls and 0/72 fault trials. A privileged inverse-and-continue witness finishes 96/96. These checks establish that selected edits matter and can be reversed; privileged state access does not test perception.

6.3. What a trajectory can and cannot show

Figure 9 separates external interventions, legal off-target placements, and terminal success in archived model traces. The extra-piece run repairs the environment's edit with another legal order. The normal run corrects its own legal but

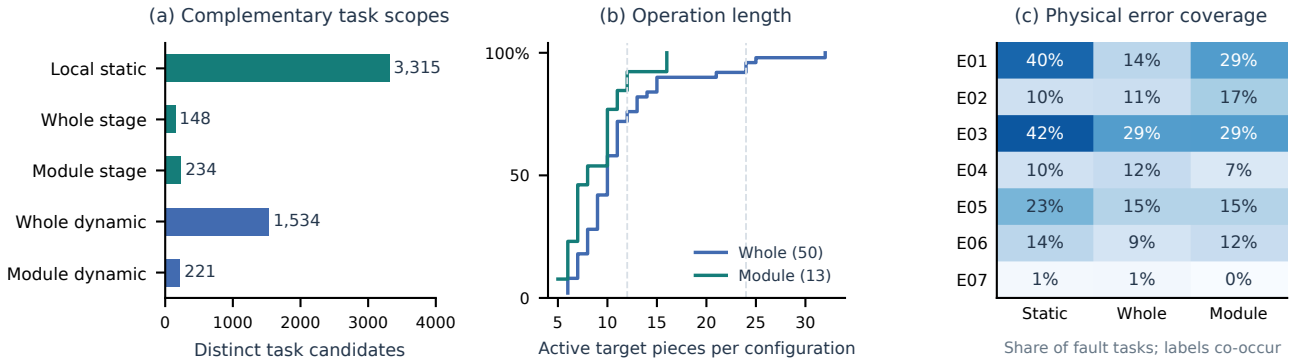


Figure 7. **Three views of benchmark coverage.** (a) Distinct main candidates by scope. (b) Target-length CDF across the 50 whole and 13 module configurations, so repeated variants do not increase an object's weight. (c) Share of faulty tasks carrying each physical label, normalized within static/whole-dynamic/module-dynamic scope; labels co-occur. These are catalogue statistics, not model performance or calibrated difficulty.

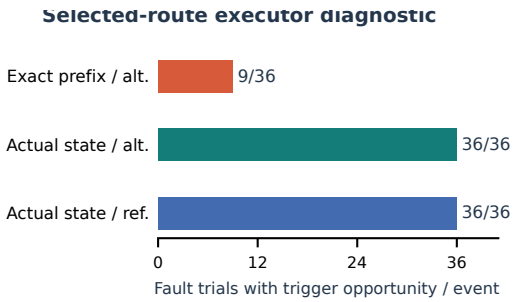


Figure 8. **Fault scheduling is itself an evaluation variable.** On the selected alternative routes, exact-reference opportunity is 9/36, while actual-state events occur in 36/36. The original twelve-module diagnostic was re-executed; the rates describe scripts, not learned-agent performance.

wrong placement. Both eventually finish, so a red intermediate state is not an episode failure. All 32 decisions replay to their recorded state hashes and feedback. These v5 development examples illustrate reporting; they are neither current-interface results nor shared-prefix causal pairs.

7. Empirical Study Design and Limitations

Comparisons and research questions. The ten retained candidate models appear in Table 3. Exact API revisions, vision support, account access, and open checkpoints must be checked before running. Use the same admitted task list, target, manual/query permissions, and effective budget for every model within a track, once per model/condition. Seed 41 fixes handle mapping. The proposed outer decision cap is $\max(40, 6N + 20)$, retaining documented native manipulation caps. Recovered requests cannot silently restart an episode to select a better answer.

Table 3. **Retained evaluation candidates.** Names are planning entries, not a claim of current endpoint availability or measured scores. Exact revisions are required before execution.

API candidates	Open visual candidates
GPT-6.1 Sol	Qwen3.5-9B; Qwen3.5-27B
Claude Sonnet 5.5	InternVL3.5-8B
Gemini 3.8 Flash	Qwen2.5-VL-7B
Claude Haiku 5.5	Qwen2.5-VL-32B
	Gemma 3-12B

The primary questions are whether normal construction predicts qualified recovery; which physical/process errors require damaging correct work; whether length, rework, or repeated incidents explain failures; and how history, manuals, views, previews, and feedback change outcomes. Each ablation fixes the remaining factors and rebuilds matched prefixes when needed. Explicit text state can expose occluded geometry and is privileged relative to RGB; free-coordinate output adds localization burden. Both require separate adapters and reports.

Admission and interpretation. Predeclare admitted task IDs and inspect their native images, inventory, and queries. Retain ambiguous cases and reasons. Adaptive-event visibility requires checking the realized location. Cost pilots record actual images, resizing, and tokens per decision; conceptual view count does not establish constant image cost.

A planned timing analysis bins incidents by physical progress $|\sigma(s) \cap \sigma(g)|/|\sigma(g)|$, preserving alternative orders. Report progress alongside B/R/T. Separate accepted moves, rejections, queries, and residual improvement before attributing errors to perception or reasoning; endpoint recovery alone does not identify its internal cause.

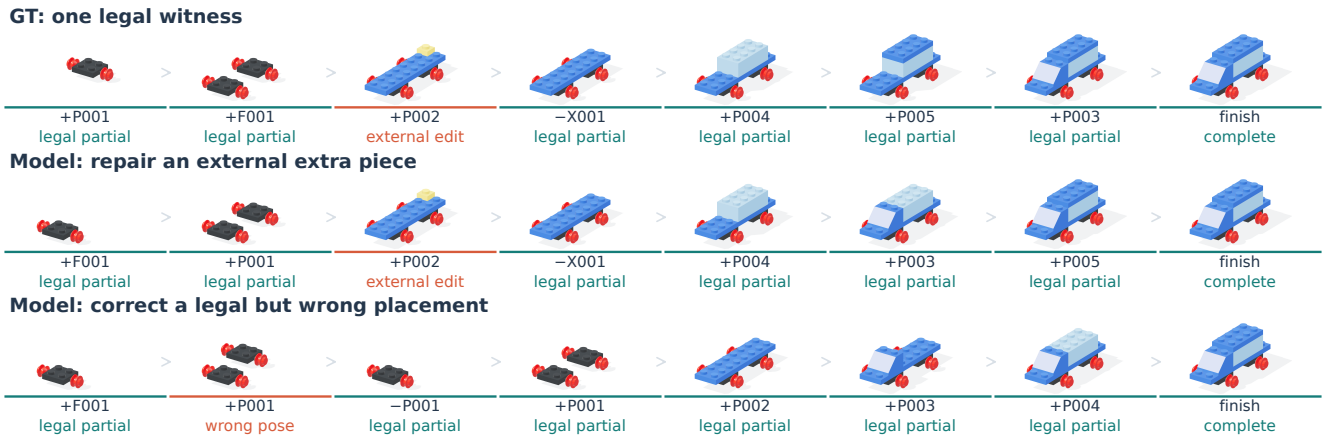


Figure 9. **GT and actual archived model repair, eight states per row.** Top: one legal reference witness. Middle: a Gemini 4 Argon rollout removes an external extra piece and completes with another order. Bottom: an independent normal rollout repairs its own legal wrong placement. All rows finish correctly. Colored rules distinguish external/off-target states from valid progress. Selected states omit inspections and intervening actions; the full 32-decision replay remains in the artifact. This archive does not populate current result tables.

Table 4. **Questions for the empirical study.** These are controlled comparisons to execute, not results. Each receives scope-specific counts, validity coverage, and the stated endpoint; the full nineteen result sheets are in the supplement.

Question	Controlled comparison	Principal outcomes	Important qualification
Construction or repair?	D0 and shared-prefix D1 branches	BCR, ICR, TR, CRR, ΔSR	Retain failed normal and untriggered trials.
Which errors require rework?	E01–E07 and P01/P02 within objects	SR, F1, ER, DR; action effort	Physical labels co-occur; witnesses are not optimal.
Do failures accumulate?	Length/rework strata and T1/T2/T3	All-event completion, window repair, retention	Partial exposure and short windows stay separate.
Which evidence matters?	History, manual, view, preview, feedback ablations	SR/BCR/CRR, queries, cost	Change one factor; state assistance is privileged.
Does action format matter?	Candidate ID versus free coordinates	Completion, schema/physics failures, cost	Coordinate adapter and matched bank are planned.
Can people read and operate it?	Native-input human panel and snapshot probe	Completion, usability, independent detection	No typed answers; no copied participant submissions.

The planned graphical human panel has 72 conditions, six participants, and three independent runs per condition, without prose or JSON. Nobody sees both pair members. Continuous trials report completion and exposure; independent participants are not cloned-history pairs. System-supplied prefixes are C tasks, and human causal rates need their own pairing design. Nineteen unmeasured result sheets in the supplement preserve static, dynamic, paired, repeated-event, input/output, detection, human, and cost analyses. Checkpoint, independent-detection, and alternative-state/coordinate banks remain to be built or integrated. No formal performance or cost estimate is inferred from these plans.

Scope of current evidence. Full-context visual admission, independent human study, paired API logging, and source-bootstrap reporting remain pending. Complete original objects, long community modules, and untouched sources are limited; public files may occur in pretraining. Finite candidates, fixed context, conservative obstacles, and straight paths simplify manipulation. Articulation, flexibility, force stability, and robot control are outside scope.

8. Conclusion

BRICKPLAN connects static correction with continuous assembly through a shared visual action contract. It accepts legal alternative routes, separates physical discrepancies from access/repetition demands, and makes exposure and control eligibility explicit. The current collection, metrics, and executor diagnostics provide a reproducible basis for the remaining learned-agent and independent human study.

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